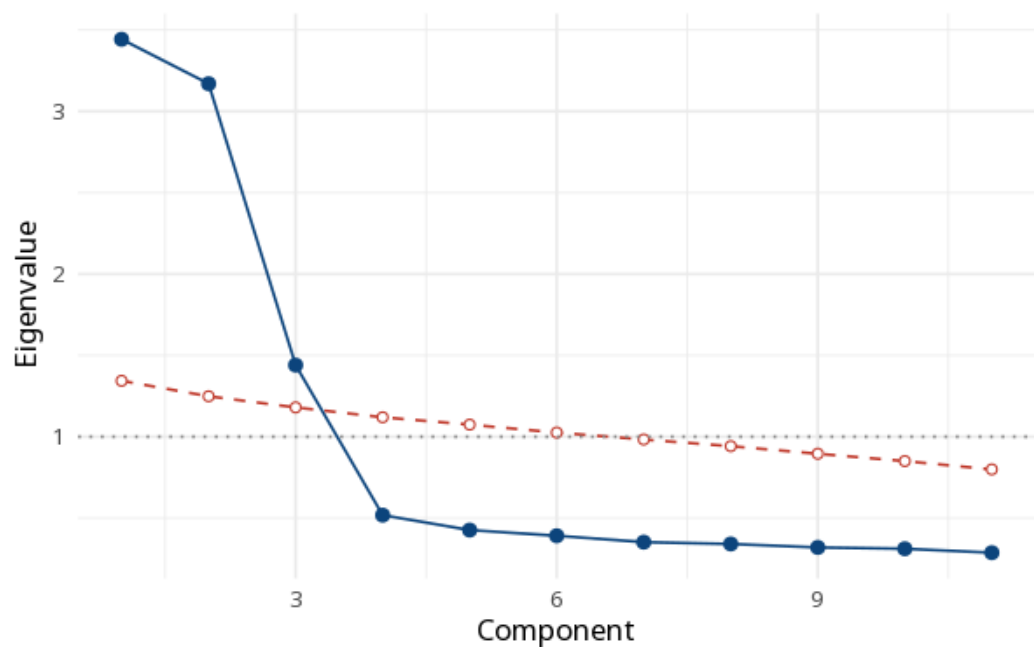


# 1 Principal component analysis (PCA)

| Component | Eigenvalue | % variance | Cumulative % |
|-----------|------------|------------|--------------|
| PC1       | 3.443      | 31.3       | 31.3         |
| PC2       | 3.170      | 28.8       | 60.1         |
| PC3       | 1.440      | 13.1       | 73.2         |

| Variable | PC1  | PC2  | PC3  |
|----------|------|------|------|
| esg1     | -.62 | -.51 |      |
| esg2     | -.56 | -.52 | .33  |
| esg3     | -.58 | -.52 | .33  |
| esg4     | -.56 | -.51 |      |
| norm1    | -.39 | .75  |      |
| norm2    | -.39 | .71  |      |
| norm3    | -.41 | .72  |      |
| norm4    | -.41 | .72  |      |
| intent1  | -.71 |      | -.53 |
| intent2  | -.66 |      | -.56 |
| intent3  | -.72 |      | -.48 |



Solid: observed · Dashed: parallel analysis 95th pct · Dotted: eigenvalue = 1

PCA compresses correlated variables into a few components. The variance explained (Table 1) shows how much information each component retains; the component loadings (Table 2) show how strongly each variable contributes.

Loadings are correlation-scaled ( $\text{eigenvector} \times \sqrt{\text{eigenvalue}}$ ) so they share the unit of a correlation coefficient. PCA is data reduction - components are weighted composites of the observed variables, not reflective latent factors; do not interpret them as constructs and do not report communalities (which belong to factor analysis, not PCA). For a latent-construct model use EFA (psych::fa) or CFA (lavaan). The scree plot overlays the parallel-analysis 95th-percentile threshold - retain components where the observed eigenvalue exceeds that line; the Kaiser eigenvalue  $> 1$  rule is shown for reference but tends to over-extract (Zwick & Velicer, 1986). Cumulative variance  $\geq 70\%$  is a common but explicitly subjective target (Jolliffe & Cadima, 2016).

PCA (correlation matrix; parallel analysis) retained 3 components explaining 73.2% of total variance.

Statistical basis: Principal component analysis as a data-reduction method (not a latent-variable model) (Jolliffe, I. T., Cadima, J. (2016). "Principal component analysis: a review and recent developments." *Philosophical Transactions of the Royal Society A*, 374, 20150202.); PCA is not EFA - components are composites, not reflective factors (Frick, S., Killisch, R., BiSSantz, N., Wetzel, E. (2025). "PCA Is Not EFA." *European Journal of Psychological Assessment*, 41(6), 425-426.); Kaiser eigenvalue  $> 1$  rule tends to over-extract (Zwick, W. R., Velicer, W. F. (1986). "Comparison of five rules for determining the number of components to retain." *Psychological Bulletin*, 99(3), 432-442.).